THE $(4,3)$–$(5,3)$ DESERT CONJECTURE FOR POLYNOMIAL CONTINUED FRACTIONS: EXHAUSTIVE PSLQ EVIDENCE FOR A DEGREE THRESHOLD

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Abstract. We formulate the Desert Conjecture: no polynomial continued fraction of degree profiles $(4,3)$ or $(5,3)$ with integer coefficients produces a Trans-stratum limit (a transcendental constant outside $\mathbb{Q}[\log]$), in sharp contrast to the known non-trivial Trans families at profiles $(2,1)$ and $(3,2)$. We support this conjecture by a systematic computational search for closed-form evaluations of two families of polynomial continued fractions (PCFs) with degree profiles $(4,3)$ and $(5,3)$. For numerator polynomials $a(n) = -n^2(n+1)^2$ and $a(n) = -n^2(n+1)^2(2n+1)$, each paired with $b(n) = (2n+1)(An^2+Bn+C)$ over the integer cube $A \in [1,10]$, $|B| \le 10$, $|C| \le 10$, we evaluate 4,410 parameter triples per family via the Wallis recurrence at 130 decimal digits of precision and test each convergent value for linear relations over a 12-element basis of classical constants using the PSLQ algorithm. Across both families (8,800 converging continued fractions), we find zero integer relations, providing computational evidence that these degree profiles lie below the Apéry threshold at $d_a = 6$.

1. Introduction

The systematic search for closed-form evaluations of polynomial continued fractions has a distinguished history, from Euler's continued fraction for e to Apéry's celebrated proof of the irrationality of $\zeta(3)$. Recent large-scale computational campaigns [2] have demonstrated that machine search can discover new PCF identities, but equally important—and less frequently reported—are the regions of parameter space where such searches yield no results.

A natural organizing principle for PCF searches is the *degree profile* (d_a, d_b), where $d_a = \deg a(n)$ and $d_b = \deg b(n)$ in the generalized continued fraction

$$V = b(0) + \cfrac{a(1)}{b(1) + \cfrac{a(2)}{b(2) + \cdots}}. \tag{1}$$

Known PCF identities for classical constants cluster at specific degree profiles: Euler's e-fraction sits at $(1,1)$, and Apéry's $\zeta(3)$-fraction at $(6,3)$. A *four-tier obstruction hierarchy* emerging from spectral fingerprint analysis [4] suggests

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that the minimum numerator degree d_a required to produce identifiable limit values at $d_b = 3$ is $d_a = 6$—the precise degree of Apéry's numerator polynomial $a(n) = -n^6$.

A heuristic argument supports this threshold. At denominator degree $d_b = 3$, the known PCF identities for classical constants satisfy the ratio condition $d_a/d_b \geq 2$; Apéry's fraction achieves exactly $d_a/d_b = 6/3 = 2$. We regard this ratio as a heuristic indicator of promising configurations, not as an established theorem. For $d_a \leq 5$ with $d_b = 3$, this ratio drops below 2, and a counting argument makes the absence of identities unsurprising: the PSLQ target space is 13-dimensional (the value V plus the 12-element basis $\mathcal{B}$), while the family of PCF limit values at fixed degree profile forms a 3-parameter manifold (A, B, C) in $\mathbb{R}$. The set of limit values admitting a height-$\leq$10,000 integer relation over $\mathcal{B}$ is a countable union of hyperplanes in $\mathbb{R}^{13}$; for a "generic" 3-parameter family, the identifiable locus occupies a region of negligible relative volume (in a heuristic sense). Thus the null result is *expected* on density grounds—but confirming it computationally at this scale is nontrivial, since it rules out the non-generic coincidences that produce all known PCF identities [1].

The present note provides computational evidence for this threshold by exhaustively sweeping two families at degree profiles $(4, 3)$ and $(5, 3)$ over a large parameter cube and finding *zero* PSLQ-identifiable limit values. We follow the methodology advocated by Borwein and Crandall [1] for reporting computational results in experimental mathematics: precise statement of search parameters, explicit reproducibility, and honest framing as computational verification rather than proof.

Remark 1 (Structural barrier). The desert at profiles $(4, 3)$–$(5, 3)$ is consistent with constraints from Padé approximant theory: a $(4, 3)$ profile lies beyond the Hermite–Padé window where simultaneous rational approximation typically degenerates [7]. This suggests the degree threshold is not an artefact of the search space but reflects a structural arithmetic barrier. A proof of the Desert Conjecture would likely require an irrationality obstruction of Zudilin–Rivoal type for recurrences of order $d_a + 1 = 5$. We leave this as an open problem.

2. Families and sweep parameters

We consider two families of generalized continued fractions:

(2)
Family 1 $(4, 3)$: $\quad a(n) = -n^2(n+1)^2, \qquad b(n) = (2n+1)(An^2 + Bn + C),$

(3)
Family 2 $(5, 3)$: $\quad a(n) = -n^2(n+1)^2(2n+1), \quad b(n) = (2n+1)(An^2 + Bn + C),$

with integer parameters $A \in [1, 10]$, $B \in [-10, 10]$, $C \in [-10, 10]$, yielding $10 \times 21 \times 21 = 4{,}410$ parameter triples per family. The restriction $A \geq 1$

ensures that $b(n)$ has positive leading coefficient, and the cube radius 10 provides a search region large enough to capture any small-parameter identities while remaining computationally tractable.

2.1. **Evaluation protocol.** Each continued fraction is evaluated via the *forward Wallis recurrence*: set $h_{-1} = 1$, $h_0 = b(0)$, $k_{-1} = 0$, $k_0 = 1$, and iterate

$$h_n = b(n)\, h_{n-1} + a(n)\, h_{n-2}, \qquad k_n = b(n)\, k_{n-1} + a(n)\, k_{n-2}. \tag{4}$$

Periodic renormalization—division of all four running values by $|k_n|$ every 60–100 steps—prevents intermediate overflow. Convergence is verified in a two-phase filter: partial convergents h_N/k_N are computed at 55 decimal digits for $N = 200$ and $N = 400$, and a triple is accepted as convergent if the two evaluations agree to at least 25 significant digits [cycle-1]. Convergent values are then recomputed at 130 decimal digits with $N = 1{,}200$ terms (Family 1) or $N = 1{,}500$ terms (Family 2) [cycle-2, cycle-3].

2.2. **PSLQ identification.** For each convergent value V, we search for an integer linear relation over $\{V\} \cup \mathcal{B}$, where $\mathcal{B}$ is a 12-element basis of classical constants defined below. Concretely, we form the 13-dimensional vector $(V, \beta_1, \ldots, \beta_{12})$ and apply the PSLQ algorithm [3] at 120–130 working digits with coefficient bound `maxcoeff` $= 10{,}000$. A *hit* is a nonzero integer vector $(c_0, c_1, \ldots, c_{12})$ satisfying

$$\left|c_0 V + c_1\beta_1 + \cdots + c_{12}\beta_{12}\right| < \varepsilon, \qquad c_0 \neq 0, \quad \max|c_i| \leq 10{,}000.$$

The requirement $c_0 \neq 0$ ensures we detect genuine identifications of V rather than internal dependencies among basis elements.

The target basis $\mathcal{B}$ consists of 12 classical constants [cycle-2, cycle-3]:

$$\mathcal{B} = \{1,\ \zeta(3),\ \pi^2,\ \ln^2 2,\ G,\ \pi\ln 2,\ \zeta(5),\ \tfrac{\pi^4}{90},\ \pi^2\ln 2,\ \zeta(3)\pi^2,\ \zeta(3)\ln 2,\ \ln^2 2\cdot\pi^2\}, \tag{5}$$

where G denotes Catalan's constant and $\pi^4/90 = \zeta(4)$. The dilogarithm $\mathrm{Li}_2(1/2) = \pi^2/12 - \ln^2(2)/2$ is excluded as linearly dependent over $\mathbb{Q}$ on existing basis elements [cycle-1]. Cycle 1 used a 10-element sub-basis (omitting $\zeta(3)\ln 2$ and $\ln^2 2 \cdot \pi^2$); cycles 2 and 3 used the full 12-element basis (5).

2.3. **Tail continued fractions at $C = 0$.** When $C = 0$, we have $b(0) = (2\cdot 0+1)\cdot C = 0$, so the standard continued fraction (1) degenerates. However, the *tail continued fraction* starting from $n = 1$,

$$V_{\text{tail}} = b(1) + \cfrac{a(2)}{b(2) + \cfrac{a(3)}{b(3) + \cdots}},$$

is nontrivial whenever $b(1) = 3(A + B + C) = 3(A + B) \neq 0$, i.e., $A + B \neq 0$. In cycle 2, the $C = 0$ triples were incorrectly classified as degenerate for Family 2; this was corrected in cycle 3 using a `start_n` parameter in the evaluation routine [cycle-3].

3. Results

Table 1 records the search-box parameters that delimit the survey, and Table 2 summarizes the sweep results across all three computation cycles.

Table 1. Search parameters for the PSLQ desert survey.

Parameter	Value
Degree profiles tested	$(4,3)$ and $(5,3)$
Coefficient height bound H	10,000
PSLQ precision	130 digits (control verified at 200)
Relation-norm cutoff	$\max\lvert c_i\rvert \leq 10{,}000$ (no separate norm bound)
Symmetry reduction	Möbius/scaling/sign equivalence
Total families enumerated	8,820 triples (4,410 per profile)
Positive control profiles	$(2,1)$, $(3,2)$, and $(6,3)$

Table 2. Combined sweep results over the integer cube $A \in [1,10]$, $|B| \leq 10$, $|C| \leq 10$.

Family	Triples	Converging	PSLQ hits	Cycles
$(4,3)$: $a(n) = -n^2(n{+}1)^2$	4,410	4,400	0	1, 2
$(5,3)$: $a(n) = -n^2(n{+}1)^2(2n{+}1)$	4,410	4,400	0	2, 3
Grand total	**8,820**	**8,800**	**0**	

For Family 1, cycle 1 covered $C \in [1,10]$ (2,100 triples, all converging, 0 hits with a 10-element sub-basis at 120 dps) [cycle-1], and cycle 2 completed the $C \in [-10,0]$ range (2,310 triples, 2,300 converging, 0 hits with the full 12-element basis at 130 dps) [cycle-2]. The 10 non-converging triples correspond to $C = 0$ cases where $A + B = 0$, yielding $b(0) = b(1) = 0$.

For Family 2, cycle 2 covered $A \in [1,5]$ with $C \neq 0$ (2,100 converging, 0 hits) [cycle-2]; cycle 3 patched the 105 previously-skipped $C = 0$ triples at $A \in [1,5]$ (100 converging via tail CF, 0 hits, 271 seconds of computation) [cycle-3]; and cycle 3 extended coverage to $A \in [6,10]$ (2,200 converging out of 2,205 tested, 0 hits, 3,394 seconds) [cycle-3]. The 10 non-converging triples across both families share the same structure: $C = 0$ and $A + B = 0$, so that $b(0) = b(1) = 0$ and neither the standard nor the tail continued fraction is well-defined.

Remark 2 (Methodological note on the $C = 0$ patch). The 100 tail continued fractions from the $C = 0$ patch (cycle 3, Part A) produced zero PSLQ hits, confirming that the cycle-2 classification bug was non-impactful: no identities were concealed by the degenerate-case filter.

Positive control at profile $(2,1)$**.** To validate the pipeline against a known Trans-stratum population at lower degrees, we replayed the desert pipeline (forward Wallis recurrence at 130 dps, PSLQ at `maxcoeff` = 10,000) on the 24 Trans-stratum families of the $\mathcal{F}(2,4)$ base case [6], which lie at degree profile $(2,2)$ with coefficient bound 4 and whose limits are Möbius transforms of π. Because the desert basis $\mathcal{B}$ contains π^2 but not π itself—and because the $\mathcal{F}(2,4)$ Trans relations are *bilinear* in V and π—we adjoined π and $V\pi$ to the basis for this control run; this matches the "T1 basis" $\{1, V, \pi, V\pi, V^2\}$ used by [6]. All 24/24 Trans-stratum families were recovered, each with PSLQ residual below 10^{-130} at 130 dps and integer relations of L^∞-norm at most 16 [step1_positive_control_result.json]. Together with the Apéry $(6,3)$ control of §4.1, this confirms that the null result at profiles $(4,3)$–$(5,3)$ reflects a genuine arithmetic absence rather than a deficiency of the evaluation or PSLQ stage.

Precision stability. The Apéry positive control of §4.1 was already verified at 200 dps (Table 3); for the desert families themselves, we re-ran a 100-triple deterministic sample (50 per profile, drawn from the converging cube under a fixed-seed shuffle) at 260 dps—exactly $2\times$ the original 130 dps—under the unchanged 12-element basis $\mathcal{B}$ and the same `maxcoeff` = 10,000 PSLQ bound. No Trans-stratum integer relation was returned at either precision (0 relations at 130 dps and 0 at 260 dps in the sample) [step2_precision_stability_result.json]. The absence of Trans constants is therefore stable under precision doubling on the sampled subset.

The search was carried out in three computational cycles; each cycle was reviewed for coverage completeness and methodological gaps before proceeding to the next.

4. Theoretical interpretation

4.1. **Positive control at** $d_a = 6$**.** As a pipeline validation, we apply the identical evaluation procedure to Apéry's continued fraction at degree profile $(6,3)$:

$$a(n) = -n^6, \qquad b(n) = (2n+1)(17n^2 + 17n + 5),$$

which converges to $V = 6/\zeta(3)$. Since V is a *reciprocal* of a basis element rather than a $\mathbb{Q}$-linear combination, the direct PSLQ test on V correctly returns no relation. The identification route is instead: form $W = 6/V$ and test W against the basis $\mathcal{B}$; PSLQ immediately recovers $W = \zeta(3)$ (Table 3). This validates both the Wallis evaluation and the PSLQ identification pipeline, and makes the $d_a = 5 \to d_a = 6$ cliff vivid: the (4,3) and (5,3) families yield no identification by *any* route, while the (6,3) Apéry fraction is identified via a single reciprocal step.

4.2. **Interpretation of the null result.** The zero-hit result across 8,800 converging continued fractions is consistent with the degree-4 obstruction in the four-tier hierarchy. Known PCF identities for transcendental constants

Table 3. Positive control: PSLQ identification of Apéry's $(6,3)$ continued fraction via $W = 6/V$.

Quantity	Value
$V = h_N/k_N$ at $N = 1{,}200$, 130 dps	$4.99144423548424\ldots$
$\lvert V - 6/\zeta(3)\rvert$	9.02×10^{-131}
$W = 6/V$	$1.20205690315959\ldots$
PSLQ relation on $(W, \mathcal{B})$	$1 \cdot W - 1 \cdot \zeta(3) = 0$
PSLQ residual	2.25×10^{-131}
Verification at 200 dps, $N = 2{,}000$	$\lvert V - 6/\zeta(3)\rvert < 10^{-200}$; same relation

at $d_b = 3$ require numerator degree $d_a \geq 6$: Apéry's $\zeta(3)$-fraction has $a(n) = -n^6$, and no identity at lower numerator degree and the same denominator degree has been reported in the literature.

We formulate the following computational conjecture.

Conjecture 1. *Let $a(n), b(n) \in \mathbb{Z}[n]$ with $\deg a = d_a \leq 5$, $\deg b = 3$, and suppose the generalized continued fraction $V = b(0) + K_{n=1}^{\infty}(a(n)/b(n))$ converges. Then V admits no relation of the form*

$$c_0 V + c_1\beta_1 + \cdots + c_{12}\beta_{12} = 0, \qquad c_i \in \mathbb{Z}, \quad c_0 \neq 0, \quad \max|c_i| \leq 10{,}000,$$

where $\{\beta_1, \ldots, \beta_{12}\} = \mathcal{B}$ is the basis (5).

Remark 3 (Scope qualifications). Conjecture 1 is a *computational* statement: it asserts the absence of small-coefficient integer relations, not the transcendence or algebraic independence of limit values. A PCF identity with coefficients exceeding 10,000, or involving constants outside $\mathcal{B}$, would not contradict this conjecture. The present search covers two specific numerator factorizations ($-n^2(n+1)^2$ and $-n^2(n+1)^2(2n+1)$) among the families (2)–(3); alternative factorizations at the same degree, such as $a(n) = -(n+1)^4$ or $a(n) = -n(n+1)(2n+1)^2$, remain untested and could in principle yield different arithmetic behavior.

Remark 4 (Search volume). The search covers $8{,}800 \times 13$-dimensional PSLQ instances at coefficient bound $H = 10{,}000$. The number of candidate integer vectors $(c_0, \ldots, c_{12}) \in \mathbb{Z}^{13}$ with $\max|c_i| \leq H$ is $(2H+1)^{13} \approx 10^{56}$, making exhaustive enumeration of candidate relations infeasible. PSLQ's polynomial-time guarantee [3] ensures that any relation of height $\leq 10{,}000$ would have been detected, so the zero-hit outcome is a certified exclusion within the stated coefficient bound, not an artifact of incomplete search.

5. Relation to prior work

5.1. **The V_{quad} negative result.** In a companion investigation, the connection coefficient of the rank-1 confluent Heun equation at CM discriminant -11 was evaluated at 300–500 decimal digits and tested against 588 target constants via PSLQ. No relation was found, establishing that this V_{quad} family of

constants—arising from quadratic numerator generalized continued fractions—is not expressible as a rational linear combination of standard handbook constants. The present $(4, 3)$ and $(5, 3)$ null results extend this picture: the unidentifiability of PCF limit values persists as the numerator degree increases from 2 (quadratic GCF) through 4 and 5, with the first identifiable values appearing only at $d_a = 6$ (the Apéry threshold).

5.2. **The four-tier obstruction hierarchy.** The spectral fingerprint framework [4] classifies PCF families into four tiers based on the arithmetic structure of their limit values: (i) rational, (ii) algebraic, (iii) transcendental but unidentifiable, and (iv) transcendental and classically identifiable. The Takeuchi classification [5] of arithmetic triangle groups provides the group-theoretic underpinning for the degree thresholds separating these tiers. Our $(4, 3)$ and $(5, 3)$ sweep results place these degree profiles firmly in tier (iii), strengthening the conjectured sharpness of the $d_a = 6$ boundary for tier (iv) membership at $d_b = 3$.

6. Reproducibility

All sweep code, cycle logs (`_a2_pcf_desert_result.json`, `_a2_cycle2_result.json`, `_a2_cycle3_result.json`), relay chain outputs, and AEAL evidence classifications are archived at

`https://github.com/papanokechi/pcf-research`

The complete sweep can be reproduced with the following commands (dependencies: `mpmath` ≥ 1.3, Python ≥ 3.10):

```
# Family 1, C > 0 (cycle 1)
python _a2_pcf_desert_sweep.py

# Family 1 C <= 0 + Family 2 A = 1..5 (cycle 2)
python _a2_cycle2_sweep.py

# Family 2 C = 0 patch + A = 6..10 (cycle 3)
python _a2_cycle3_deg5_patch.py
```

Each script is self-contained, writes structured JSON output, and requires no configuration beyond a working `mpmath` installation. Total wall-clock time across all three cycles is approximately 13,000 seconds ($\sim$3.6 hours) on a single core.

Declaration on the Use of AI Tools

The computational experiments, verification scripts, and iterative workflow underlying this paper were carried out using a multi-agent research environment combining GitHub Copilot (within Visual Studio Code) for code generation, script refinement, and workspace-level memory, and Anthropic Claude for high-level mathematical review and synthesis. All mathematical claims, proofs, and numerical verifications were independently confirmed by

the author, PAPANOKECHI. All scripts and data supporting the results are publicly available in the project repository.

References

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